

Escherichia coli Mutant Strain with Altered Expression of the Tryptophan Operon: Ribonucleic Acid Synthesis in Vitro

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Ribonucleic acid (RNA) synthesis has been studied in vitro with a partially purified preparation of RNA polymerase from a mutant strain of *Escherichia coli* with a reduced rate of accumulation of tryptophan RNA (P. H. Pouwels and H. J. Scholten, J. Bacteriol. 139:393-397, 1979). The incorporation of radioactive label into RNA with polymerase from mutant bacteria is considerably lower than that with the enzyme from wild-type bacteria. These results are explained by the presence in mutant bacteria, but not in wild-type bacteria, of a factor which suppresses the accumulation of RNA. Mutant bacteria contain a factor which renders RNA synthesis with mutant and wild-type RNA polymerase resistant to various inhibitors of RNA synthesis, e.g. rifampin, streptolydigin, and heparin. We conclude that in mutant bacteria a factor is modified which suppresses the accumulation of some RNA species and lowers the sensitivity of RNA polymerase to some transcription inhibitors.

In the accompanying paper (23) we described the isolation and preliminary characterization of a strain of *Escherichia coli* with a reduced rate of synthesis of tryptophan (*trp*) mRNA. In mutant bacteria the rate of incorporation of [³H]uracil into pulse-labeled RNA, but not stable RNA, is reduced.

To gain more insight into the nature of the mutation, we studied RNA synthesis in vitro with partially purified RNA polymerase preparations from mutant and wild-type bacteria. Our results indicate that mutant bacteria, but not wild-type bacteria, contain a factor which suppresses the accumulation of RNA by purified RNA polymerase. In addition, mutant bacteria contain a factor which renders RNA polymerase partially resistant to various inhibitors of RNA synthesis.

MATERIALS AND METHODS

Bacteria. The bacterial strains (TL105 Trp⁺ Lac⁺ and TL105 Trp⁻ Lac⁻) used in this study have been described in the accompanying paper (23).

Purification of RNA polymerase. Except when stated otherwise, transcription studies were carried out with a partially purified preparation of RNA polymerase (DE enzyme) from wild-type or mutant bacteria, which was obtained in the following way. Bacteria were cultivated at 34°C in broth and were harvested in the exponential phase of growth. A cellular extract was prepared as described by Burgess and Jendrisak (3). After centrifugation (2 h at 40,000 rpm) to remove the particulate fraction, RNA polymerase was purified by (NH₄)₂SO₄ fractionation (33 to 50%

saturation), followed by chromatography on DEAE-cellulose, as described by Burgess (2). The fractions from the DEAE-cellulose column containing RNA polymerase activity (DE enzyme) were pooled, dialyzed against 10 mM Tris (pH 7.9), 0.1 mM dithiothreitol, 0.1 mM EDTA, 0.2 M NaCl, and 50% glycerol (storage buffer), and stored at -20°C. Alternatively, DE enzyme was further purified by chromatography on Ultrogel (ACA 22) equilibrated with 10 mM Tris (pH 7.9), 0.1 mM EDTA, 0.1 mM dithiothreitol, 1 M NaCl, and 5% glycerol. The active fractions from the Ultrogel column (ACA enzyme) were pooled, dialyzed against storage buffer, and stored at -20°C.

RNA synthesis in vitro. RNA synthesis was carried out at 32°C for 10 min in a reaction mixture (50 µl) containing: T7 DNA (40 µg/ml) or ϕ80trpEA190 DNA (27 µg/ml), 40 mM Tris-acetate (pH 7.9), 1 mM dithiothreitol, 0.4 mM of CTP, ATP, and GTP, 0.05 to 0.4 mM [³H]UTP (125 to 200 Ci/mol), 10 mM magnesium acetate, 50 mM KCl, and 1 mg of bovine serum albumin per ml. After addition of the enzyme, RNA synthesis was started by placing the reaction mixture in a 32°C water bath and was stopped by cooling the reaction mixture in ice, after addition of 1 ml of carrier RNA (1 mg/ml) and 1 ml of cold trichloroacetic acid (10%). The precipitate that formed was collected on Whatman glass-fiber filters and washed with cold acid and ethanol, and the radioactivity present on the filters was determined with a liquid scintillation counter.

In experiments in which the effect of rifampin on the activity of binary complexes of DNA and RNA polymerase was studied, T7 DNA and enzyme were incubated at 32°C in a reaction mixture lacking the nucleoside triphosphates. After incubation for 10 min a mixture of CTP, ATP, and GTP (0.4 mM each), 0.1 mM [³H]UTP, and rifampin (final concentration 0 to

250 $\mu\text{g/ml}$) was added. RNA synthesis was for 3 min, and the amount of RNA synthesized was assayed as described above. To determine the sensitivity of DNA-enzyme complexes to heparin, complexes were formed by incubating DNA in the presence of the enzyme for 10 min at 32°C. Subsequently heparin was added (0.1 mg/ml), and RNA synthesis was started after various periods of time by taking a sample and adding a mixture of ribonucleoside triphosphates. RNA synthesis was carried out at 32°C and was stopped after 3 min, and the amount of RNA synthesized was determined as described above.

RESULTS

RNA synthesis in vitro. In the accompanying paper (23) we presented evidence that the mutation causing the reduced rate of synthesis of *trp* enzymes in a mutant strain of *E. coli* (TL105 Trp⁻ Lac⁻) is caused by a defect in the accumulation of RNA. To gain more insight into the possible defect in RNA accumulation, we studied the rate of RNA synthesis in vitro. When RNA polymerase was partially purified by chromatography on DEAE-cellulose, a protein fraction resulted (DE enzyme) which was significantly more active when prepared from wild-type than from mutant bacteria (Fig. 1). To determine whether the decreased rate of incorporation of [³H]UMP into RNA with the enzyme preparation from mutant bacteria is a result of a reduced rate of synthesis of RNA or of an increased rate of degradation of RNA we compared the rate of degradation of ³H-labeled T7 RNA with DE enzyme preparations from wild-type and mutant bacteria. [³H]RNA was synthesized with T7 DNA as a template, and after 8 min rifampin was added to stop further initiation of RNA synthesis. Subsequently the release of radioactivity into acid-soluble material was determined. Our results (not shown) show that, despite the presence of rifampin, some residual incorporation into RNA takes place with the enzyme preparation from mutant, but not wild-type, bacteria. This surprising result prompted us to investigate in more detail the effect of rifampin on RNA synthesis in vitro.

Figure 2 shows the result of an experiment in which the effect of two concentrations (0.3 $\mu\text{g/ml}$ and 10 $\mu\text{g/ml}$) of rifampin on the rate of RNA synthesis was measured with DE enzyme from wild-type (Fig. 2A) and mutant bacteria (Fig. 2B). The results clearly show that, in contrast to wild-type RNA polymerase, the mutant enzyme is partially resistant to rifampin. At the lower concentration (0.3 $\mu\text{g/ml}$) of rifampin tested, RNA synthesis with mutant enzyme was only slightly affected, but with the higher concentration (10 $\mu\text{g/ml}$) a considerable reduction of the rate of RNA synthesis was found.

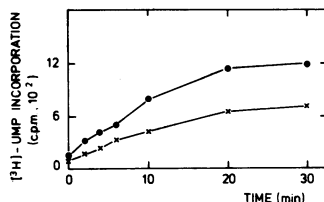


FIG. 1. RNA synthesis in vitro on $\phi 80trpEA190$ DNA (27 $\mu\text{g/ml}$) with partially purified RNA polymerase (DE enzyme; 0.40 mg/ml each) from wild-type (○) and mutant (×) bacteria. The specific activity of [³H]UTP (50 μM) was 200 Ci/mol.

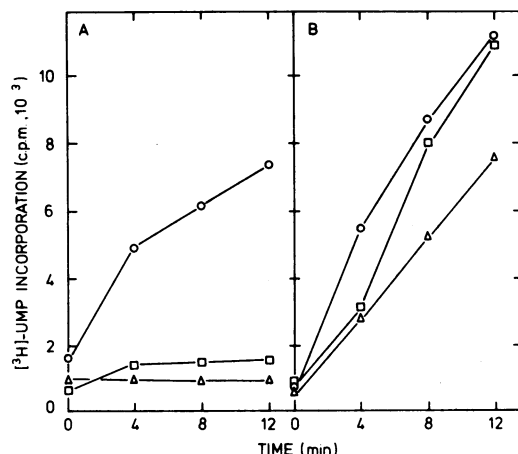


FIG. 2. RNA synthesis in vitro on T7 DNA (40 $\mu\text{g/ml}$) in the presence of rifampin with partially purified RNA polymerase (DE enzyme) from wild-type (A) and mutant (B) bacteria. The specific activity of [³H]UTP (0.4 mM) was 125 Ci/mol. (○) Without rifampin; (□) 0.3 μg of rifampin per ml; (Δ) 10 μg of rifampin per ml.

Mutant bacteria contain a factor which suppresses the accumulation of RNA by purified RNA polymerase. The results described could be explained by assuming that RNA polymerase in mutant bacteria is modified, which renders the enzyme partially resistant to rifampin. Alternatively, a factor might be present in mutant but not in wild-type bacteria, causing the resistance of RNA polymerase to this drug.

To discriminate between these possibilities we took two approaches. First, we attempted to determine the sensitivity to rifampin of mixtures of purified wild-type RNA polymerase and partially purified mutant RNA polymerase. Second, we further purified RNA polymerase from mutant bacteria in an attempt to separate the hypothetical factor from the RNA polymerase.

Prior to the determination of the sensitivity to rifampin of a mixture of mutant enzyme and

wild-type enzyme, we measured the activity of mixtures of partially purified RNA polymerase from mutant and wild-type bacteria and highly purified wild-type enzyme. When a fixed amount of wild-type DE enzyme was mixed with increasing amount of purified wild-type enzyme, the activities found were the sum of the two activities separately, as expected. However, the activities of the same amount of mutant DE enzyme and increasing amounts of purified wild-type RNA polymerase were not additive. In contrast, it was found that mutant DE enzyme strongly suppresses the activity of wild-type enzyme, suggesting that mutant DE enzyme, but not wild-type DE enzyme, contains an activity which reduces the accumulation of RNA by RNA polymerase. The result of this experiment made less meaningful an attempt to determine the effect of mutant DE enzyme on the sensitivity of wild-type enzyme to rifampin.

Separation from RNA polymerase of a factor which renders RNA polymerase resistant to rifampin. We also further purified an RNA DE polymerase preparation from mutant bacteria by gel filtration on an agarose column. RNA polymerase was separated from the bulk of the other proteins since it is eluted ahead of most other proteins. As a result of this purification step the resistance of the enzyme to rifampin was lost. Addition of the fractions that were eluted from the agarose column after the RNA polymerase activity (and which are essentially free of RNA polymerase activity), however, considerably restored resistance to rifampin (Table 1). When these fractions were added to wild-type RNA polymerase, a similar degree of resistance to the antibiotic was found. A similar amount of material obtained from wild-type bacteria, which had been purified in an identical fashion, did not affect the resistance of RNA polymerase to rifampin (Table 1). From our results we conclude that in mutant bacteria, but not in wild-type bacteria, a factor is present which renders RNA polymerase resistant to rifampin. Interestingly, the fraction from mutant bacteria that renders RNA polymerase resistant to rifampin also suppressed the activity of mutant and wild-type RNA polymerase (compare lines 3 and 4, Table 1). In contrast, the same fraction from wild-type bacteria had a stimulating effect on the activity of RNA polymerase. Although the extent of the suppression of RNA polymerase activity caused by the mutant fraction varied from one experiment to another and was affected by the concentration of RNA polymerase added, the effect on the sensitivity of the enzyme to rifampin was not affected by the conditions of the experiment.

The sensitivity of mutant DE enzyme to streptolydigin and heparin. To determine whether the mutation specifically affects sensitivity to rifampin we also studied the effect of two other inhibitors of RNA synthesis, streptolydigin and heparin. Like rifampin, these drugs inhibit RNA synthesis by interaction with the enzyme, although their mode of action is quite different: rifampin and heparin interfere with the formation of an initiation complex, but streptolydigin inhibits the polymerization reaction (4, 14, 21).

Our results (Fig. 3) show that the DE enzyme from mutant bacteria is more resistant to streptolydigin than the DE enzyme from wild-type bacteria. It should be noted, however, that, as with rifampin, the mutant enzyme is only partially resistant to the drug.

The inactivation by heparin of the mutant

TABLE 1. Effect of a fraction from mutant bacteria on the sensitivity to rifampin of RNA polymerase

Purified RNA polymerase ^a	Agarose fraction added ^b	RNA synthesized		
		- Rifampin (cpm)	+ Rifampin (cpm)	Percent of control
Mutant enzyme	None	56,692	8,276	15
	Mutant	44,468	25,220	57
Wild-type enzyme	None	112,525	6,300	6
	Mutant	38,121	21,886	57
	Wild type	326,997	13,220	4

^a RNA synthesis was performed as described in the text in the absence or presence of rifampin (0.3 µg/ml) with purified RNA polymerase (ACA enzyme) from mutant or wild-type bacteria.

^b Fractions eluted from an agarose (Ultrogel) column after the RNA polymerase activity were added where indicated.

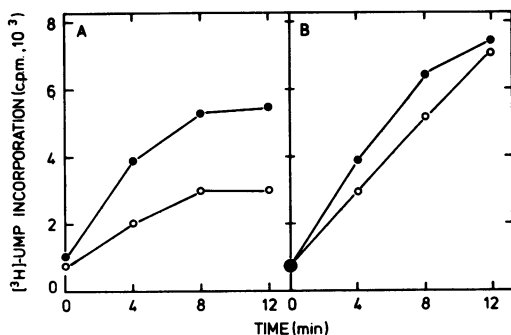


FIG. 3. RNA synthesis in vitro on T7 DNA (40 µg/ml) in the presence of streptolydigin with partially purified RNA polymerase (DE enzyme; 1.0 mg/ml) from wild-type (A) and mutant (B) bacteria. (●) Without streptolydigin; (○) 25 µg of streptolydigin per ml.

and wild-type enzymes was studied by measuring the rate of transcription of initiation complexes between T7 DNA and DE enzyme, incubated in the presence of heparin. After various periods of time samples were taken, and RNA synthesis was initiated by the addition of a mixture of the four ribonucleoside triphosphates. The rate of RNA synthesis was determined by assaying the amount of RNA synthesized 3 min after addition of the ribonucleoside triphosphates, a period which is short enough to prevent reinitiation of RNA synthesis in the absence of the drug. The results (Fig. 4) show that the rate of inactivation of binary complexes of DNA and enzyme is threefold faster for wild-type than for mutant enzyme.

In similar experiments we studied the kinetics of inactivation by rifampin of mutant and wild-type DE enzyme by measuring the rate of transcription of initiation complexes, when challenged with a mixture of the four ribonucleoside triphosphates and an increasing amount of rifampin. As with heparin, the rate of inactivation of binary complexes of DNA and enzyme was threefold faster for wild-type than for mutant enzyme (results not shown).

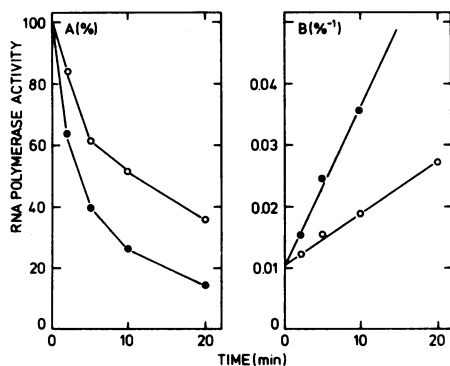


FIG. 4. (A) Kinetics of inactivation by heparin of binary complexes of T7 DNA and partially purified RNA polymerase (DE enzyme) from wild-type (●) and mutant (○) enzymes. Binary complexes between T7 DNA and DE enzyme were formed as described in the text. After addition of heparin (100 μ g/ml), samples were taken at the indicated times and added to a mixture of the four ribonucleoside triphosphates to start RNA synthesis. The results are presented as the percentage of the amount of RNA synthesized (in 3 min) in the absence of heparin (100% value). The 100% values are: wild-type enzyme, 12,177 cpm; mutant enzyme, 17,518 cpm. The amount of mutant enzyme was two to three times that of wild-type enzyme. (B). Data from (A) replotted as the reciprocal value of the percentage of the enzymatic activity versus the time of inactivation of the binary complex with heparin.

DISCUSSION

The pleiotropic character of the bacterial mutant, described in the accompanying paper, might be due to more than one genetic lesion. This should be kept in mind when considering the two major results of this paper: (i) the accumulation of RNA by RNA polymerase preparations from mutant bacteria is reduced, and (ii) RNA polymerase preparations from mutant bacteria are resistant to inhibitors of RNA synthesis.

The kinetics of inactivation of RNA polymerase-DNA complexes by heparin or rifampin have shown that the mutant initiation complex is less sensitive to the drugs than its wild-type counterpart. This difference in sensitivity could be explained by assuming that the rate of chain initiation is faster with mutant than with wild-type DE enzyme. Alternatively, the mutant initiation complex is inactivated less rapidly by the drug than the wild-type initiation complex (14, 21). Since the rate of RNA synthesis in the absence of drugs with mutant DE enzyme is slower than with wild-type DE enzyme, both *in vivo* and *in vitro*, the former possibility seems highly unlikely. In addition, the finding (Fig. 3) that RNA synthesis with mutant DE enzyme is less sensitive than that with wild-type DE enzyme to the inhibitor of transcription elongation, streptolydigin, argues in favor of a difference in rate of inactivation of mutant and wild-type enzyme. Another explanation for the difference in sensitivity of mutant and wild-type RNA polymerase activity to the inhibitors of RNA synthesis, i.e., that the inhibitors are inactivated by the mutant factor, seems highly unlikely, in view of the fact that the three inhibitors are structurally quite different.

Although the mutant DE enzyme is inactivated more slowly than the wild-type DE enzyme, it is not fully resistant to rifampin, as is the enzyme from rifampin-resistant bacteria. It is therefore not surprising that mutant bacteria are sensitive to rifampin (Pouwels, unpublished data). Moreover, the mutant strain used in our experiments was not selected for by its resistance to rifampin, as are rifampin-resistant bacteria.

The reduced rate of incorporation of radioactive label into RNA with mutant RNA polymerase preparations and the resistance of the enzyme to rifampin can be explained by the presence of two activities in mutant, but not in wild-type, bacteria. One activity suppresses the rate of incorporation of radioactive label into RNA, and another activity renders RNA polymerase resistant to rifampin. The two activities can be separated from RNA polymerase. Since both activities are present in the same fractions of the

Ultragel column in the final purification step, they may reside in the same factor.

RNA polymerase is rapidly inactivated by rifampin because it binds to the enzyme, thereby preventing the initiation of RNA synthesis (12, 27). RNA polymerase isolated from rifampin-resistant strains is no longer inhibited by the antibiotic (25). Resistant strains arise by a mutation mapping at a single position of the *E. coli* chromosome (1). The resistance of the mutant enzyme has been shown to reside in the β subunit (13).

The factor that renders RNA polymerase resistant to rifampin could be one of the subunits of the enzyme. To explain the resistance of the mutant enzyme by a mutation in the β subunit, we would have to assume that β (in part) is dissociated from the enzyme in the Ultragel purification step. In addition we would have to assume that mutant β displaces its counterpart on the wild-type enzyme, since the factor renders wild-type enzyme resistant to rifampin. We consider this possibility unlikely, since the subunits of RNA polymerase are dissociated under very stringent conditions only (28). Another, more likely explanation would involve a mutation in the subunit σ . Rifampin binding studies with core and holoenzyme have indicated that σ affects the rifampin binding site on the enzyme (26). Mutations in σ which affect the interaction with core enzyme, therefore, may alter the sensitivity to inhibitors of the enzyme. A mutation in σ , β , or both may also explain the reduction of the rate of RNA synthesis *in vitro* by the factor isolated from mutant bacteria.

The factor that renders RNA polymerase resistant to rifampin also may be a cellular component that is not a subunit of the enzyme. For example, mutant bacteria may contain an altered factor involved in transcription or translation. Several proteins which affect transcription have been isolated and characterized (7-9, 16, 19, 20). An alteration of such a protein might have an effect both on the rate of transcription and on the sensitivity of the enzyme to inhibitors.

Genetic (5) and biochemical (6, 10, 11) evidence suggests that transcription and translation are coupled (24). The accumulation of some mRNA species is selectively shut off *in vivo* (15, 18) when protein synthesis is inhibited. In addition it has been shown that tRNA, which plays a key role in protein synthesis, can directly interact with RNA polymerase (22). It is conceivable that a mutation in one of the components of the translation machinery diminishes the sensitivity of RNA polymerase to inhibition by rifampin. Such a mutation might also explain the

reduced rate of RNA accumulation with RNA polymerase preparations from mutant bacteria.

The reduced rate of incorporation of radioactive label into RNA by RNA polymerase preparations from mutant bacteria could be explained by a reduced rate of synthesis or by the presence of an RNase with increased activity. The latter hypothesis would require that the RNase is highly specific, since in mutant bacteria the incorporation of radioactive label into RNA is reduced for some mRNA species only. An altered RNase would not easily explain the resistance of RNA polymerase to inhibitors like rifampin.

Based on the arguments outlined above, a possibly oversimplified model that can be advanced to account for the results described in this and the accompanying paper would be that in mutant bacteria a factor is altered which changes the conformation of RNA polymerase. This conformational change would result in the diminished capacity of the enzyme to synthesize some species of mRNA and also in a decrease of the sensitivity of the enzyme to rifampin. To test the correctness of the model will require a detailed genetic analysis of the mutant strain.

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